Project: ATABA REMEDIATION ANALYTICS - 2025 LAND USE/LAND COVER AND CHANGE DETECTION STUDIES

INTRODUCTION:

Ataba community was impacted by two (2) major spills from Bodo in 2008 for which the community secured a judgement in court against SPDC (The Shell Petroleum Development Company, now Renaissance Africa Energy Company Limited). SPDC and Ataba community subsequently signed a MoU to guide the two years cleanup and remediation of 550Ha of impacted land in Ataba in November 2020. (Source: Ogoni Restoration Team).

A baseline land use land cover mapping was done in 2021 using a high-resolution satellite imagery acquired in 2022, to provide basis for trend analysis and change detection required for monitoring vegetation (Mangrove) regrowth in Ataba.

This report contains the statistical analysis of the land use/land cover classification result of the satellite imagery over the Ataba area of interest acquired on the 18th of December 2024 to highlight the current area extent of the vegetation cover as at the time of satellite data capture. The Area of interest (AOI) was advised by the ORP team and has been consistently used for all previous and present studies.

STUDY OBJECTIVES:

The objectives of this study are:

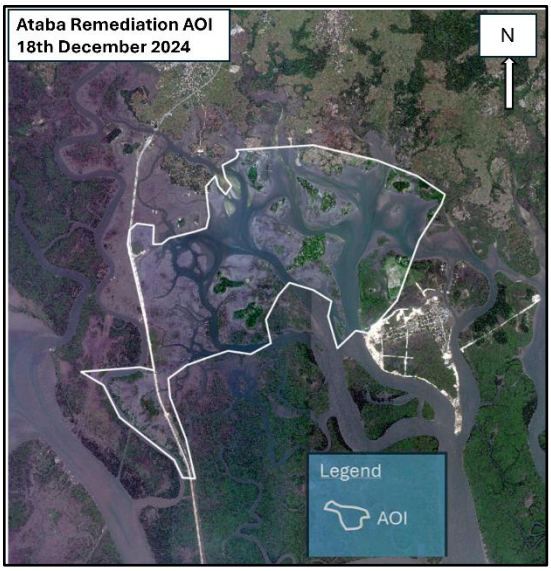
- To identify, map, extract and evaluate area covered by vegetation.
- To identify evidence of vegetation regrowth within Ataba AOI and determine the differences in the status of land use and land cover over 2 epochs within the Ataba AOI from 2024 to 2025.
- To determine the different variations and estimate the progress of mangrove regrowth within the Ataba AOI.

The methodology to achieve these objectives include Image compositing and classification using the Support Vector Machine (SVM) algorithm & geometry extraction calculations and data analytics.

STUDY AREA:

The study area covers Ataba communities within the Oil Mining Lease (OML 11) in the Niger Delta. The combined area of the entire area of study is approximately 31.53 km^2.

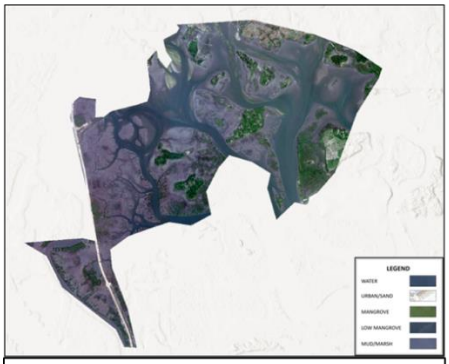
The terrain is mainly mangrove swamps with creeks and tidal or mudflats, the study area is susceptible to tidal actions. The study area is accessible via land and rivers, and the climate is tropical monsoon.



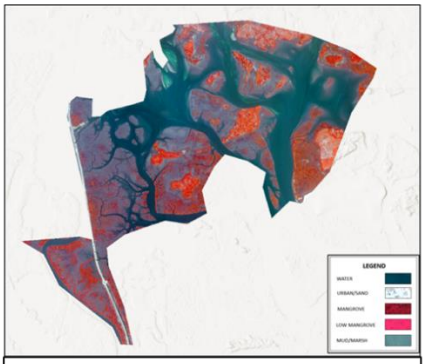
METHOLOGY AND RESULTS:

The satellite data was first corrected for the attenuating effect of atmospheric moisture, dust particles and fine clouds (atmospheric correction) and then geometrically corrected (georeferencing/coregistering) . A 'False Colour' (FC) image was then generated from the corrected data through a process called band compositing using bands combination of band-4 (Near-Infra red), band-3 (Red) and band-2 (Green) in that order in a RGB colour compositing model. On the false colour image, the very intense red colour indicate dense, vigorously growing vegetation. As plant vigor decreases, the vegetation appears as lighter shades of red and pink, the blue, grey and white hues indicate water, mud/marsh and built-up areas (both categorize as Water and Others in the 2024 classification result).

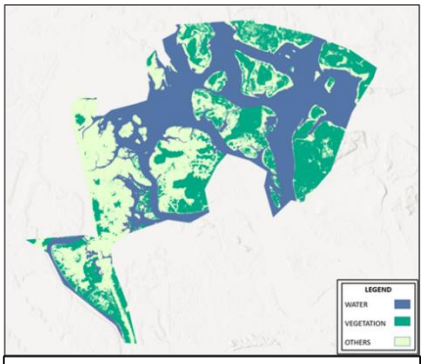
The satellite image data (18th December 2024 for 2025 study year) was georeferenced with the January 2024 satellite image and classified using the support vector machine classification method in ArcGIS Pro to generate land use and land cover class distribution for the area of interest. The baseline information for each class was extracted from the False Colour Composites (FC) combination of the RGB raster data and a statistical analysis of the identified feature classes was done to quantify the area of each land use/land cover class.



True Colour Composite 18th December 2024 (For 2025 Study)

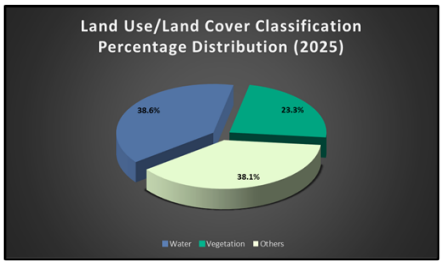


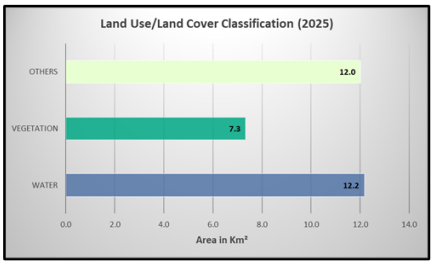
False Colour Composite 18th December 2024 (For 2025 Study)



Land use/Land cover Classification 18th December 2024 (For 2025 Study)

Epoch: 2025		
Land Use/Land Cover Types	**Area (Km2)**	**Percentage (%)**
Water	12.2	38.6
Vegetation	7.3	23.3
Others	12.0	38.1
TOTAL	31.54	100





LANDUSE AND LAND COVER ANALYSIS:

The False Colour Composite (FC) image clearly distinguished between the built-up areas and sand features (Roads and bare soil), and vegetation areas within the area of interest. The vegetated areas clearly stood out in red, in comparison to their bare and partly submerged surroundings (i.e., mud/marsh areas). Shallow tidal areas and deeper channels of the creeks within the area of interest were also clearly distinguishable. The result shows that water covers 38.6%, vegetation covers 23.3% and others cover 38.1% of the entire AOI.

Water:

Water class increased from 9.8 km² in January 2024 to 12.2 km² in December 2024. This increase in water level may be attributed to the changes in the tide as at the time of image capture.

Vegetation

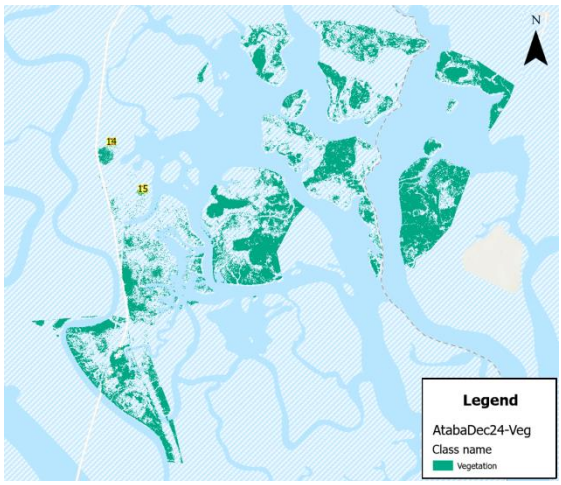
Vegetation conspicuously reduced from 9.5 km² in January 2024 to 7.3 km² in December 2024. This reduction is probably due to evident pockets and large mangrove vegetation loss distributed across the Ataba area of interest. From visual interpretation, the vegetation loss may be attributed to environmental/natural phenomena or human activities and interference.

Others:

No significant change observed. The others class slightly reduced from 12.2 km² to 12.0 km².

Comment: The analysis of Ataba satellite imagery shows consistent results with other imageries over the Ataba remediation AOI till date. It also shows a clear view of the time lapse in the mangrove distribution over time. Further analysis shows the evidence of vegetation recovery in the AOI.

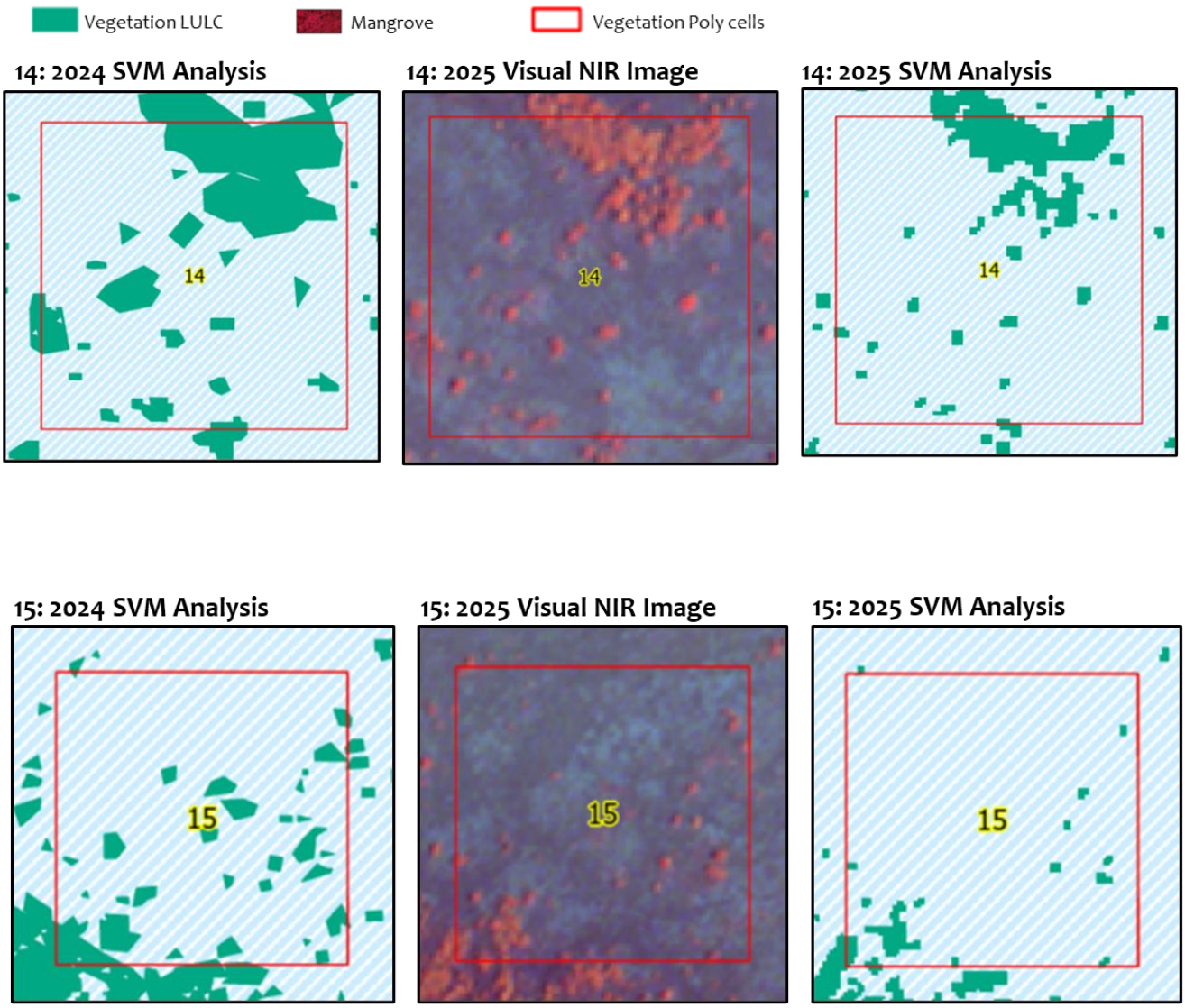
CHANGE ANALYSIS FOR MANGROVE RECOVERY WITHIN ATABA AOI (2022 & 2025)



The methodology adapted for this study is a change detection analysis to identify evidence of vegetation regrowth in Ataba AOI and determine the differences in the status of land use and land cover over 2 epochs within the Ataba AOI from 2022 to 2025.

To determine the different variations and estimate the progress of mangrove regrowth within the Ataba AOI, a total of two (2) cell sites were identified to help measure each mangrove presence visually. These 17 cells were provided by the ORP team. These measurements were then used as a basis for comparing the image processing results for those areas. The two (2) cell sample locations were selected to have various levels of mangrove coverage and were roughly distributed within the area of interest.

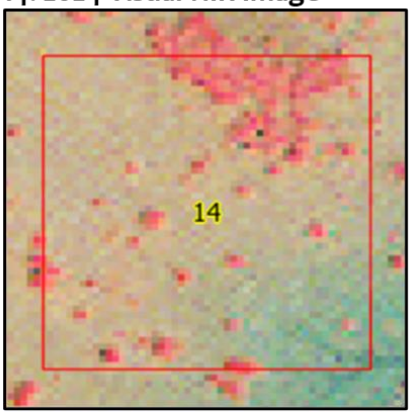
Below are the interpretation and classification results for the January 2024 SVM analysis and the December 2024 SVM analysis using the two (2) cells vegetation polygon outline data integrated with the December 2024 Ataba NIR raster data. The cells have been categorized as cells 14 and 15, respectively. Margin of error between January 2024 SVM data and the December 2024 SVM data classification is + or – 5m.



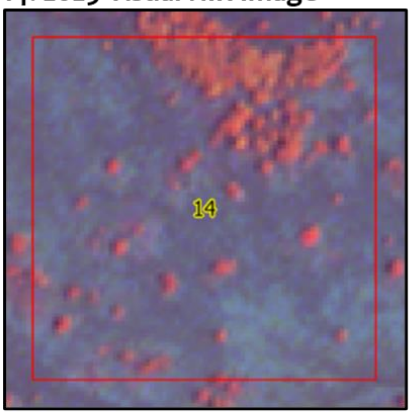
Comparison of Classification results for January 2024 and December 2024 using SVM algorithm

Change Detection Analysis Between January 2024 Visual NIR and December 2024 (For 2025 Study Year) Visual NIR Images (Cells 14 & 15)

14: 2024 Visual NIR Image

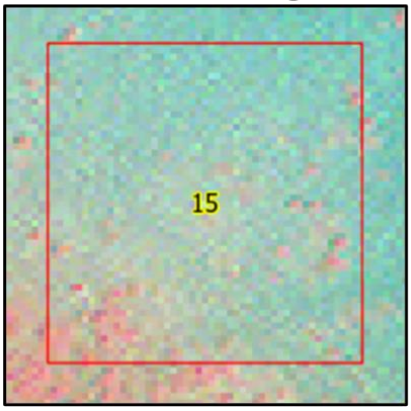


14: 2025 Visual NIR Image

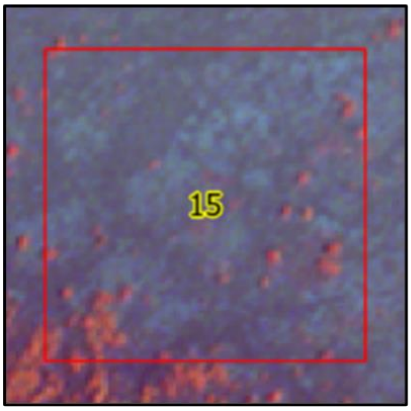


14 - The distribution and pattern of vegetation appear similar between both year of study. There are no significant growth or difference observed between both images.

15: 2024 Visual NIR Image



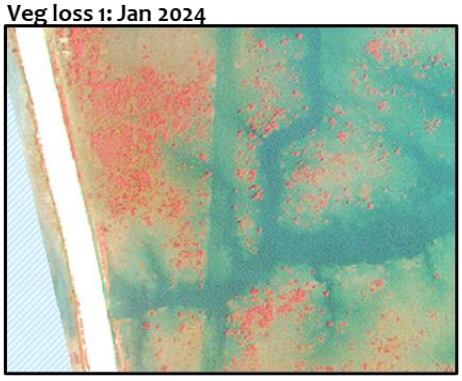
15: 2025 Visual NIR Image



15 - There are no significant mangrove regrowth observed between both images. The distribution and pattern of vegetation appear similar between both year of study.

Change Detection Analysis Between January 2024 Visual NIR and December 2024 visual NIR (Cells 14 and 15).

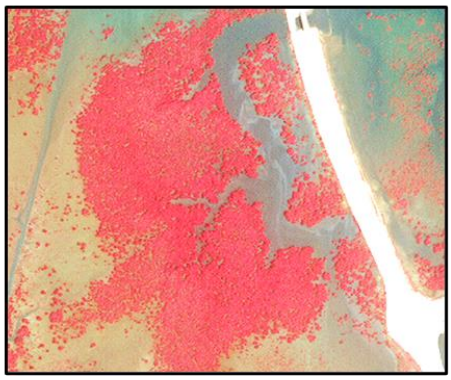
Veg loss 1: Jan 2024



Veg loss 1: Dec 2024

Veg Loss 1 – Significant vegetation loss observed in area circled in yellow. Vegetation loss most likely due to human or natural interference.

Veg loss 2: Jan 2024



Veg loss 2: Dec 2024

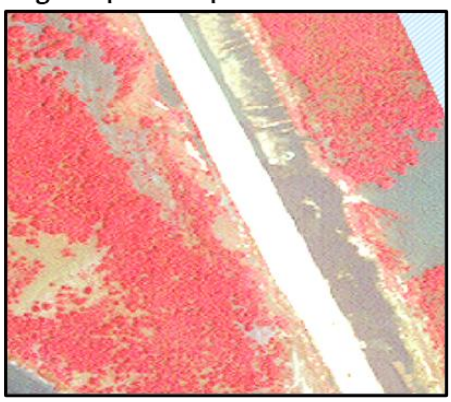
Veg Loss 2 – Pockets of vegetation loss observed in area circled in yellow. This is most likely due to human or natural interference.

Veg loss 3: Jan 2024

Veg loss 3: Dec 2024

Veg Loss 3 – Significant vegetation loss observed in area circled in yellow. Vegetation loss most likely due to human or natural interference.

Veg loss 4: Jan 2024

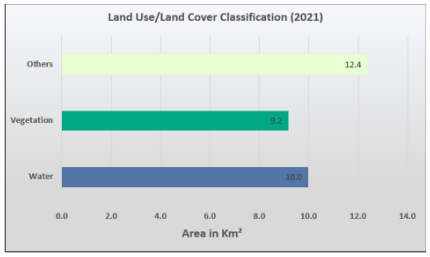


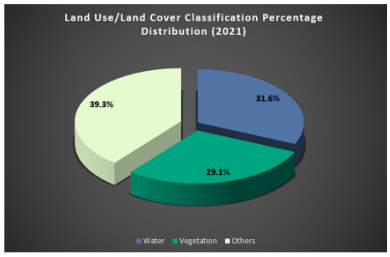
Veg loss 4: Dec 2024

Veg Loss 4 – Pockets of vegetation loss observed in area circled in yellow. This is most likely due to human or natural interference.

2021

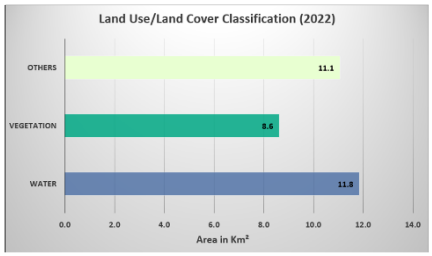
Epoch: 2021		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	10.0	31.6
Vegetation	9.2	29.1
Others	12.4	39.3
TOTAL	31.54	100

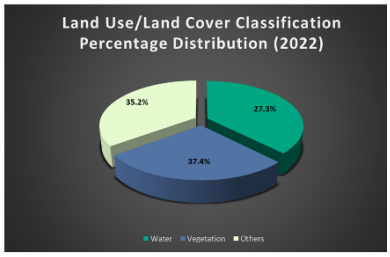




2022

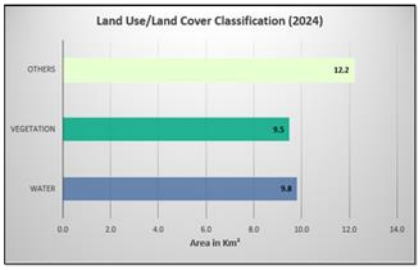
Epoch: 2022		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	11.8	37.4
Vegetation	8.6	27.3
Others	11.1	35.2
TOTAL	31.53	100

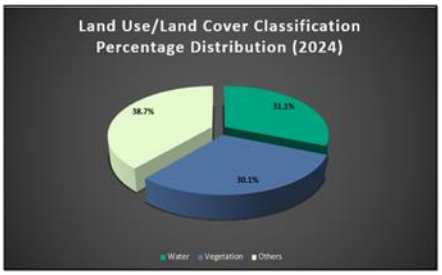




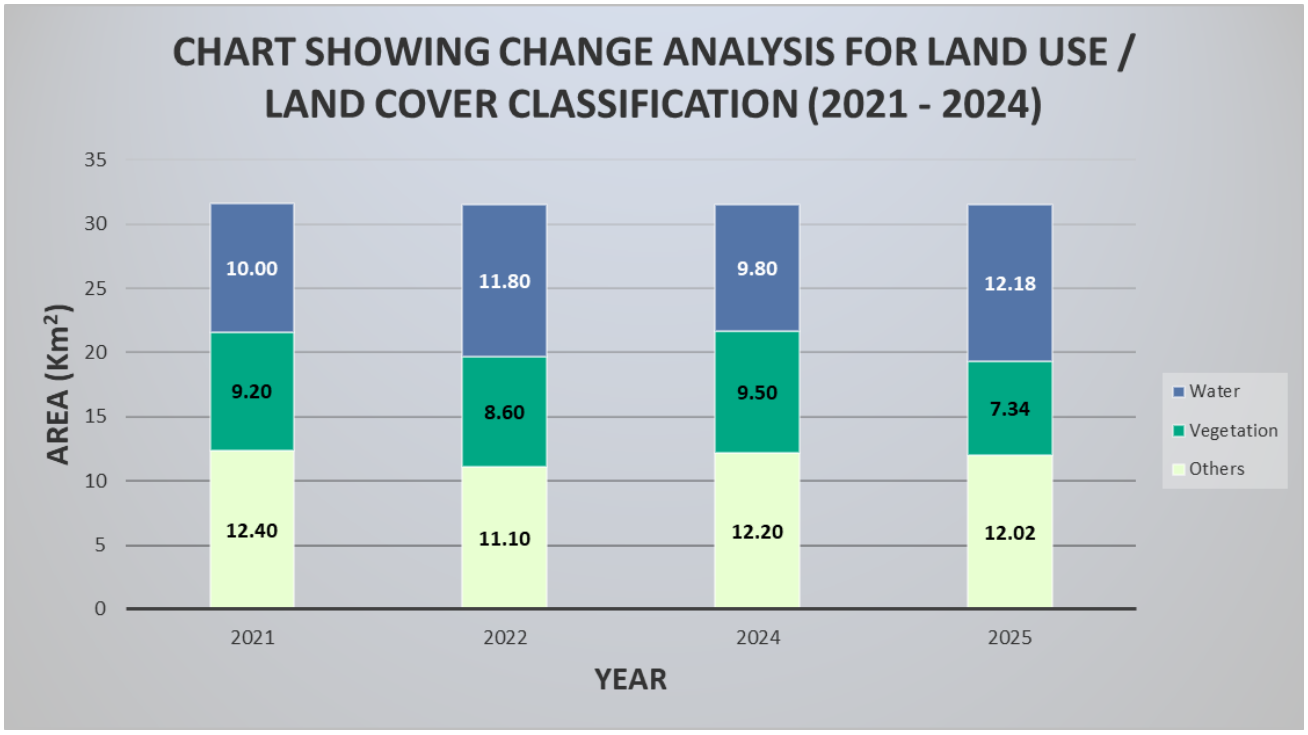
2024

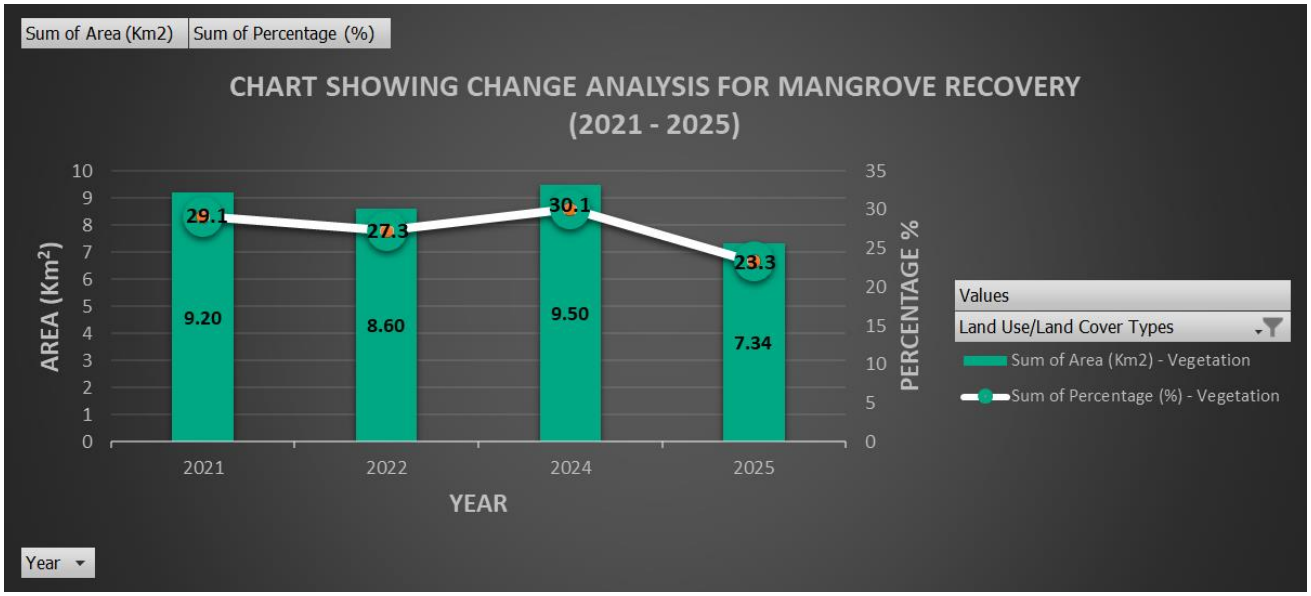
Epoch: 2024		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	9.8	31.1
Vegetation	9.5	30.1
Others	12.2	38.7
TOTAL	31.53	100



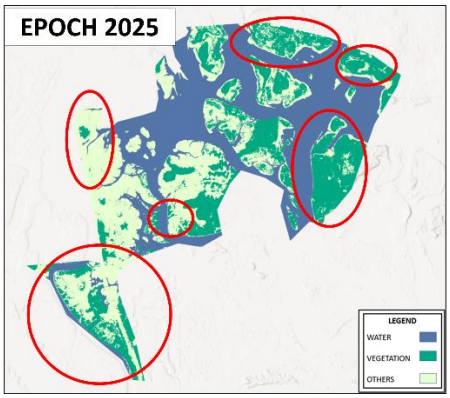


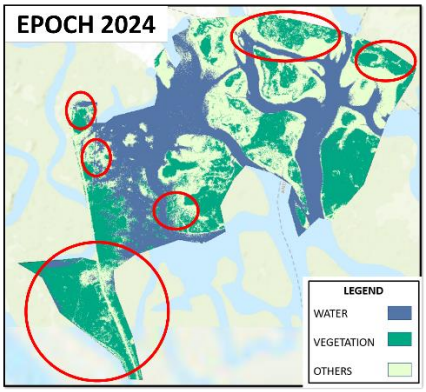
Ataba classification output and statistical analysis (2021 – 2024)

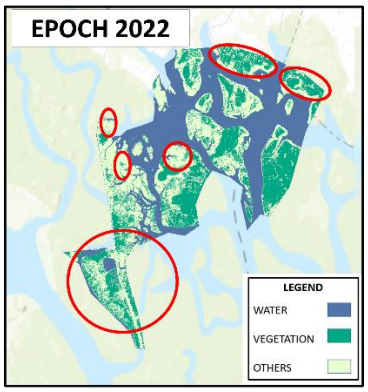


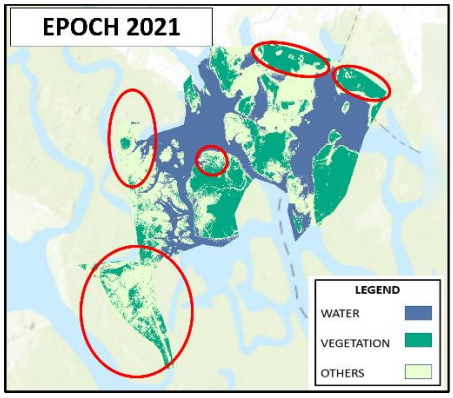


Charts showing change analysis for land use/land cover studies.









Change analysis for mangrove within Ataba AOI

Year	Land Use/Land Cover Types	Area (Km2)	Percentage (%)
2021	Water	10.0	31.6
2021	Vegetation	9.2	29.1
2021	Others	12.4	39.3
2022	Water	11.8	37.4
2022	Vegetation	8.6	27.3
2022	Others	11.1	35.2
2024	Water	9.8	31.1
2024	Vegetation	9.5	30.1
2024	Others	12.2	38.7
2025	Water	12.2	38.6
2025	Vegetation	7.3	23.3
2025	Others	12.0	38.1

Area and Percentage of The Land Use / Land Cover Classes Within Ataba Aoi From 2021 – 2025

LULC	2024 AREA (sqr km)	2025 AREA (sqr km)	AREA CHANGED (sqr km)	STATUS
Water	9.80	12.18	2.38	Gain
Vegetation	9.50	7.34	2.16	Loss
Others	12.20	12.02	0.18	Loss
TOTAL	31.5	31.5		

Land Use / Land Cover Net Gain/Loss Change Between 2024 - 2025

CONCLUSIONS & RECOMMENDATIONS:

CONCLUSIONS:

The study successfully deployed remote sensing and geospatial technologies to map, extract and estimate area of vegetation cover, water, and others, which was subsequently used for a time-lapse change analysis for mangrove regrowth in the area of study between year 2021 and year 2025, enabling the Ogoni Restoration Project (ORP) team monitor, plan and manage their replanting campaigns and other related activities.

The result of the statistical analysis for the 2025 classification study shows a land cover/land use percentage area distribution:

Water:

Water class increased from 9.8 km^2 in January 2024 to 12.2 km^2 in December 2024. This increase in water level may be attributed to the changes in the tide as at the time of image capture.

Vegetation

Vegetation conspicuously reduced from 9.5 km^2 in January 2024 to 7.3 km^2 in December 2024. This reduction is probably due to evident pockets and large mangrove vegetation loss distributed across the Ataba area of interest. From visual interpretation, the vegetation loss may be attributed to environmental/natural phenomena or human activities and interference. More effort is required in cells 14 and 15 to increase the vegetation of these areas as very little changes were noticeable in the result of the analysis.

Others:

No significant change observed. The others class slightly reduced from 12.2 km^2 to 12.0 km^2.

The result of this analysis is to feed into a more robust analysis of all available satellite imagery over the Bodo remediation area of interest till date to have a clear view of the time-lapse in mangrove distribution using Support Vector Machine' classification method in combination with a segmented image, adequate training data sets and field verification of results.

RECOMMENDATIONS:

1. Actual status of these sites (cells 14 and 15) should be verified during field visits.
2. We suggest that more cell site locations within the Ataba study area where subsequent replanting will be done should be shared by the Ogoni Restoration Project Team to enable a more focused monitoring and better analysis and interpretation of Remote sensing data.

*A story map is presently being created and can be accessed via this Ataba, Bodo Reforestation Project link and will be live on the 20th of October, 2025.

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